

Risk factors associated with secondary enucleation after fractionated stereotactic radiotherapy in uveal melanoma

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ABSTRACT.

Purpose: To evaluate risk factors for secondary enucleation after fractionated stereotactic radiotherapy (fSRT) in uveal melanoma.

Methods: In this retrospective study, clinical data of 118 consecutive patients who had initially been treated with fSRT between 1999 and 2009 were collected and analysed. The patients who had undergone secondary enucleation were identified and examined for clinical, histopathological and cytogenetical (fluorescence *in situ* hybridization determined) data. Also, the reasons for secondary enucleation, such as treatment failure (progressive tumour growth or tumour recurrence) or complications following fSRT (painful blind eye), were recorded and examined.

Results: The secondary enucleation rate was 16% after a median follow-up of 4.7 years, with 5% due to treatment failure and 11% due to complications. In the univariate analysis, large tumour diameter ($p = 0.019$) and large tumour height ($p = 0.001$) were associated with secondary enucleation, tumour involvement of the optic disc showed borderline significance ($p = 0.068$). Cox regression multivariate analysis displayed large tumour height as independent prognostic factor (HR 1.42, 95% CI 1.12–1.81, $p = 0.004$). Following secondary enucleation, mitotic figures were present in five of 18 tumours, and gain of chromosome 8q was also present in five tumours. Within the subgroup of patients who required secondary enucleation due to failed tumour control by fSRT ($N = 6$), mitotic figures were present in four of six tumours while gain of 8q was present in three of six tumours.

Conclusion: Secondary enucleation after previous fSRT was associated with large tumour height. High mitotic counts and gain of chromosome 8q were frequently found in secondary enucleations and possibly indicate a more aggressive or radiation-resistant tumour.

Key words: enucleation – genetics – radiotherapy – uveal melanoma

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Introduction

Uveal melanoma (UM) is the most frequent intra-ocular malignancy among adults. Local therapy is based on achieving adequate tumour control with – if possible – conservation of the eye, vision and cosmetic appearance. Nowadays, most small- and medium-sized UMs are treated with one of the available eye-conserving therapies with excellent tumour control rates (Aziz et al. 2009; Gambrell et al. 2007). Brachytherapy and proton beam radiotherapy are frequently used techniques as well as stereotactic radiotherapy. However, proton beam radiotherapy is an expensive technique and has limited availability across the world. Stereotactic radiotherapy is a suitable alternative to proton beam radiotherapy, providing comparable tumour control rates especially when administered in several treatment sessions as with fractionated stereotactic radiotherapy (fSRT; Muller et al. 2005; Krema et al. 2009; Dunavoelgyi et al. 2011).

Despite the success of fSRT, 11–24% (Jampol et al. 2002; Damato & Lecuona 2004; Aziz et al. 2009) of the patients treated with eye-conserving techniques developed complications or were resilient to local tumour control requiring secondary enucleation or

reirradiation (Damato & Lecuona 2004; Marucci et al. 2011). Little is known about risk factors for secondary enucleation following fSRT. The aims of this study, therefore, were to evaluate what tumour characteristics – clinical, histopathological and genetical – predispose to secondary enucleation after fSRT. Fluorescence *in situ* hybridization (FISH) analysis was performed to identify which cytogenetic alterations were present in the secondary enucleation cases.

Materials and Methods

All consecutive patients diagnosed with choroidal or ciliary body melanoma who had been treated with fSRT between 1999 and 2009 were included in this retrospective study. A diagnosis of UM was established by indirect ophthalmoscopy, fundus photography and ultrasonography. Standardized B-scan ocular ultrasonography was used for the determination of largest basal tumour diameter and tumour height. Clinical data such as gender, age at time of diagnosis, tumour characteristics and co-morbidities were collected and recorded at baseline. All tumours were scored according to the 7th edition of TNM tumour classification (Kivelä & Kujala 2013) as well as the Collaborative Ocular Melanoma Study (COMS)-classification (Boldt et al. 2008). A total dose of 50 Gy in five fractions of 10 Gy was delivered on five consecutive days. The Rotterdam eye fixation system and the treatment techniques *per se* have been described in detail in a previous publication (Muller et al. 2005). The study was performed according to guidelines of the Declaration of Helsinki, and informed consent was obtained from all patients prior to therapy.

The patients were evaluated 6 weeks after irradiation, at 3 months and every 3 months thereafter for the first 2 years. After 2 years of follow-up, patients were evaluated at four monthly intervals. During these visits, response to treatment was evaluated by fundoscopy and ultrasound measurement of intra-ocular tumour dimensions, and patients were screened for metastases by liver enzyme blood tests. Abdominal ultrasonography was conducted every 6 months or immediately after detection of elevated blood liver enzymes. In the case of ultrasono-

graphical abnormalities, abdominal computed tomography (CT) or magnetic resonance imaging (MRI) scanning was conducted. If complications requiring therapy were found, these were treated accordingly.

Secondary enucleation was performed if there was failure of local tumour control following fSRT, that is progressive tumour growth or tumour recurrence. Another reason for secondary enucleation to be performed was if a complication from fSRT occurred, that is painful blind eye due to intractable neovascular glaucoma or in combination with radiation retinopathy. Progressive tumour growth was determined whether there was intra-ocular tumour growth of more than 25% on two or more follow-up visits. Tumour recurrence was determined whether there was tumour growth on two or more follow-up visits after a period of evident tumour regression or no visible tumour tissue remaining. Clinical parameters such as patients' and tumour characteristics were analysed at baseline before the administration of fSRT. The histopathological and cytogenetical parameters could only be examined after secondary enucleation was conducted and therefore represent post-fSRT characteristics. The patients who underwent secondary enucleation were also evaluated according to our standard follow-up protocol as mentioned before. Further follow-up data regarding development of metastasis and tumour-related death were obtained from medical records and by contacting the general physician.

Pathologic research

Conventional histopathological examination was performed on all secondarily enucleated eyes and confirmed the origin and type of the tumour, as well as tumour dimensions. Cell type was defined and recorded, as well as the presence of extravascular matrix patterns, mitotic figures, necrosis, scleral invasion, optic nerve invasion and neovascular membranes of the iris. For extracellular matrix patterns, closed loop networks were evaluated and defined as at least three back-to-back loops. Furthermore, these were evaluated using non-counterstained PAS stain without using a dark green filter. Mitotic figures were counted in an equivalent of 50 high power fields

(HPF) and viewed under the microscope at 400× magnification with a single field view of 0.45 mm in diameter. This related to a total area of 7.95 mm².

Cytogenetic research

Fresh tumour tissue was analysed for presence of chromosomal alterations by FISH (chromosomes 1p, centromere 3, 3q, 6p, 6q, 8p, centromere 8 and 8q) as described by Naus et al. (2002) and Mensink et al. (2009). For direct FISH, 1 ml of the cell suspension was fixed and used, while in seven cases, insufficient fresh tumour tissue was available to test multiple loci and FISH was carried out on paraffin sections of 4–5 µm, which were pretreated by dewaxing with xylene, permeabilizing with sodium thiocyanate, proteolysis and denaturation. The concentration for centromere probes was 5 ng per slide, whereas for the BAC clones, 15–25 ng per slide was used. After washing and staining, slides were counterstained with 4',6-diamidino-2-phenylindole and mounted in antifade solution (Dabco-Vectashield 1:1; Vector Laboratories, Burlingame, CA, USA). In all cases, signals were counted in 300 interphase nuclei, according to the criteria of Hopman et al. (1988). Cut-off threshold for deletion on fresh tumour tissue were >15% of the nuclei with one signal and for gain >10% of the nuclei with three or more signals, as described by van Dekken et al. (1990). The cut-off threshold for deletion on paraffin sections (>25% of the nuclei with one signal) was adapted from our own research as a measure to correct for truncation and cutting artefacts. The cut-off threshold of 10% for gain was left unchanged as truncation, and cutting artefacts are not a major issue for cells showing more than two signals.

Statistical analysis

Univariate Log-rank analysis or Cox proportional hazard analysis was performed for the identification of significant clinical variables predicting secondary enucleation. Cox regression multivariate analysis was performed with all significant variables in univariate analysis and was used to identify the independent value of the prognostic factors. All p-values were considered

significant at a two-tailed p-value of <0.05. A Kaplan–Meier graph was constructed of time to secondary enucleation. The analyses were constructed using the statistical software program SPSS Version 22.0 (SPSS Inc., Chicago, IL, USA).

Results

In our study, a total of 118 patients had received fSRT with a median follow-up of 4.7 years ranging to 11.0 years (118 eyes were treated). Demographics and clinical features of the patients are summarized in Table 1. Two patients were lost to follow-up before the end of our study: they had been followed-up for 57 and 81 months, respectively, till the ages of 82 and 73 years, without undergoing secondary enucleation and without signs of metastasis at that moment. Univariate analysis of the

single prognostic factors showed a significantly higher risk of secondary enucleation for patients with large tumour diameter and large tumour height, while tumours which involved the optic disc showed borderline significance. Cox regression multivariate analysis displayed large tumour height (HR 1.42, 95% CI 1.12–1.81, $p = 0.004$) as independent significant prognostic factor for secondary enucleation after fSRT (Table 2).

Of all patients treated with fSRT, 19 patients (16%) had to undergo secondary enucleation during follow-up: six of them due to failure of local tumour control (5%) (four had suffered progressive intra-ocular tumour growth and two tumour recurrence) and 13 (11%) due to complications following fSRT (12 had painful blind eye due to neovascular glaucoma and one due to diffuse radiation retinopathy) (Table 3).

One of the eyes that had to be enucleated because of complications following fSRT contained massive inflammation with necrotic cell debris and no vital tumour cells remaining (case 19). Histopathological and cytogenetical research of tumour tissue from this eye was therefore not possible. The remaining 18 eyes all yielded sufficient tumour material – fresh or paraffin embedded – for successful research of histopathological and genetical factors. Patients who underwent secondary enucleation due to a painful blind eye after fSRT (case 7–19) all had tumours with a height >5 mm, except for one patient who had a tumour measuring 4.7 mm in height (case 15). Mitotic figures (≥ 5 per 50 HPF) were present in five tumours; four had been enucleated because of failure of tumour control by previous fSRT (case 2,3,4,5) and one experienced complications from fSRT (case 7).

Loss of chromosome 3, or monosomy 3, was found in 10 tumours (56%), and simultaneous gain of chromosome 8q was present in three tumours (case 5, 10 and 16). Gain of chromosome 8q was present in five tumours (28%), three of these tumours were enucleated due to failed tumour control and two due to a painful blind eye after fSRT. In the subgroup of eyes that had been enucleated because of failed tumour control (case 1–6), three of six tumours (50%) had gain of 8q which was found in at least 20% of tumour cells analysed from one tumour (case 2), ranging to 83% in another (case 4). In the subgroup that had been enucleated due to painful blind eye (case 7–19), there were two tumours with gain of 8q; one in 18% of cells (paraffin embedded tissue) where only 60 cells could be counted due to low quality of tumour cells (case 10), and the other in 18% of cells (fresh tissue) where the normal amount of 300 nuclei had been counted (case 16).

Gain of a complete copy of chromosome 8 was found in another five tumours: three of them in the failed tumour control subgroup and the remaining two in the complications subgroup. All of these had gain of chromosome 8 in more than 23% of tumour cells, with an exception in the two tumours from the complications group: the first had gain of chromosome 8 in 30% of nuclei as well as gain of chromosomes 1p, 6p, 6q, but with

Table 1. Demographics and clinical features of patients treated by fSRT ($N = 118$).

	Mean and median	Range	p-Value*
Age at diagnosis	62 and 62 years	28–84 years	0.77
Largest basal tumour diameter	12.0 and 11.9 mm	7.2–18.9 mm	0.019
Tumour height	6.2 and 6.2 mm	2.2–11.1 mm	0.001
	Number	Percentage	p-Value†
Gender			
Female	54	46	0.082
Male	64	54	
TNM classification			
T1	21	18	
T2	48	40	
T3	47	40	
T4	2	2	
TNM staging			
I	21	18	
II	48	40	
III	49	42	
IV	0	0	
COMS classification			
Medium-sized tumour	103	87	
Large-sized tumour	15	13	
Involvement of ciliary body			
No	111	94	0.84
Yes	7	6	
Posterior margin			
<3 mm from optic disc/macula	82	70	0.56
>3 mm from optic disc/macula	36	30	
Optic disc involvement			
No	105	89	0.068
Yes	13	11	
Secondary enucleation			
No	99	84	
Yes	19	16	

TNM classification, 7th Edition.

COMS classification, Boldt et al. (2008).

* Cox proportional hazard analysis.

† Univariate log-rank test.

Table 2. Clinical markers for secondary enucleation in 118 fSRT-treated patients.

Variable	Hazard ratio	95% Confidence interval	p-Value*
Tumour height	1.42	1.12–1.81	0.004
Largest basal tumour diameter	1.05	0.87–1.28	0.61

* Cox regression multivariate analysis.

loss of chromosome 3 and therefore possibly represented a hypertriploid tumour with relative loss of chromosome 3 (case 12), the latter had gain of chromosome 8 in 85% of nuclei with chromosomal gain of all other probes tested, thereby possibly representing a hypertriploid case without relative loss of chromosome 3 (case 15). There were no additional extra copies of chromosome 8q found in this study other than one extra copy.

All above-mentioned cases and their probability of secondary enucleation-free survival are depicted in Fig. 1. Melanoma-related metastasis was eventually found in 25 of the total of

118 patients (25.3%), and five of them had also undergone secondary enucleation (26.3%). The two patients from the treatment failure subgroup, who developed metastasis, had a disease-free interval of 26 and 36 months, while this interval was 57, 98 and 67 months for the patients in the complications subgroup.

Discussion

Sixteen per cent of the patients required secondary enucleation after initial radiotherapy, which is comparable to other studies: Dunavoelgyi et al. (2011) and Fernandes et al. (2011); report rates

of around 16–27% for secondary enucleation following fSRT. While Aziz et al. (2009); Egan et al. (1998); Fuss et al. (2001); Gragoudas et al. (2002) and Macdonald et al. (2011) report 11–24% following proton beam radiotherapy, Furdova et al. (2014) report 12% after one step LINAC-based stereotactic radiosurgery and around 11% for conservative therapies combined (Damato & Lecuona 2004) all with a 5–15 years follow-up. It is somewhat difficult, however, to compare studies because of variation of inclusion criteria for tumour size and location, as well as follow-up time, the type of conservative therapy, therapeutic doses delivered, the number of fractions used and the strategy for managing painful neovascular glaucoma.

Our results show that the overall risk of secondary enucleation following fSRT increased significantly if tumours had a large height. Large tumour height is a known prognostic

Table 3. Clinical, histopathological and genetical features of patients who had undergone secondary enucleation.

Clinic			Pathology										Cytogenetics					Reason for secondary enucleation		
N	Sex	Age	Largest tumour diameter (mm)	Tumour height (mm)	Time to metastasis (months)	Time to sec. enucleation (months)	Optic disc involved	Mixed or epitheloid cell type	Extravascular matrix patterns	> 5 Mitotic figures / 50 HPF	Ciliary body involved	Neovascular membrane	Scleral invasion bij tumour	Loss of chromosome 1p	Loss of chromosome 3	Chromosome 6	Gain of chromosome 8		Gain of chromosome 8q	Tumour tissue
1	F	73	15.0	11.1		3		x							x	−6q	x		P	Progressive tumour growth
2	F	66	11.4	2.2		85				x						+6p		x	P	Tumour recurrence
3	M	80	18.9	8.4	26	17	x	x	x	x	x			x	x	−6q	x		F	Progressive tumour growth
4	M	84	13.5	5.6	36	15				x					x	+6pq	x		P	Progressive tumour growth
5	F	67	13.1	4.8		41				x				n/a	x	+6p−6q		x	F	Tumour recurrence
6	M	44	14.5	8.8		64	x	x										x	F	Progressive tumour growth
7	M	67	11.6	9.9	57	14		x		x		x		x	x				F	Painful blind eye (NVG)
8	M	45	15.2	7.8	98	55		x				x	x			+6p			P	Painful blind eye (NVG)
9	M	51	10.7	6.9		45		x	x			x	x	x	x	n/a			F	Painful blind eye (NVG)
10	M	48	12.3	9.9		17		x				x	x		x	n/a		x	P	Painful blind eye (NVG)
11	M	62	10.2	10.5		43	x	x				x		x	x				F	Painful blind eye (NVG)
12	M	28	12.3	7.3		43	x	x				x		+1p	x	+6pq	x		F	Painful blind eye (NVG)
13	M	65	8.8	5.9		32						x		x					F	Painful blind eye (NVG)
14	F	38	16.0	7.8		56										−6p−6q			P	Painful blind eye (radiation retinopathy)
15	M	73	15.2	4.7		22	x					x		+1p	+3	+6pq	x		F	Painful blind eye (NVG)
16	M	77	11.1	7.1		18		x	x			x	x	x	x	+6p		x	F	Painful blind eye (NVG)
17	M	44	9.3	10.5		15		x				x	x			−6p			P	Painful blind eye (NVG)
18	F	56	12.4	10.1		24			x			x	x			n/a			F	Painful blind eye (NVG)
19	M	74	13.4	8.5	67	9								n/a	n/a	n/a	n/a	n/a	n/a	Painful blind eye (NVG)

>5 Mitotic figures per 50 high power fields present.

NVG = neovascular glaucoma, n/a = data not available.

Tumour tissue: Paraffin embedded (P); Fresh tumour tissue (F)

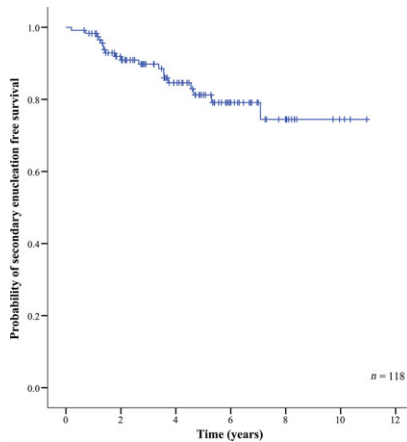


Fig. 1. Kaplan–Meier analysis of secondary enucleation-free survival rate ‘time to secondary enucleation’ in 118 fSRT-treated patients.

factor for secondary enucleation (Zehetmayer et al. 2000; Gragoudas et al. 2002; Jampol et al. 2002; Damato & Lecuona 2004), as is involvement of the optic disc by tumour (Fuss et al. 2001; Gragoudas et al. 2002; Damato & Lecuona 2004), location close to the fovea (Gragoudas et al. 2002; Jampol et al. 2002) and large tumour diameter (Fuss et al. 2001; Gragoudas et al. 2002; Damato & Lecuona 2004). Painful neovascular glaucoma is also a known cause for secondary enucleation after radiotherapy: Dunavoelgyi et al. (2013) reported 20% of all patients treated by hypofractionated stereotactic radiotherapy, to develop severe neovascular glaucoma which was the greatest risk for secondary eye loss. Neovascular glaucoma in this case was also associated with large tumour height in general (Puusaari et al. 2004; Muller et al. 2011) and more specifically, a height >5 mm (Damato 2004, 2006) and posterior location of the tumour (Dunavoelgyi et al. 2013). However, nearly all enucleated eyes and especially the cases from the complications subgroup also received extensive medical treatment to the affected eye before and after fSRT which may have increased the risk of secondary enucleation as well. It is unknown which histopathological and genetical alterations in the tumours could have been induced or altered by the fSRT. Toivonen et al. (2003) demonstrated more necrosis and lower extravascular matrix patterns in irradiated, secondarily enucleated tumours compared to primarily enucleated tumours. Comparison of tumour tissue acquired by biopsy

before and after radiation therapy could therefore provide us with essential information. Several groups already reported fine-needle biopsy to be a safe and reliable technique yielding sufficient tumour tissue for cytogenetic analysis (Naus et al. 2002; Midenas et al. 2006; Shields et al. 2011; Abi-Ayad et al. 2013).

In our study, we found four tumours with increased mitotic figures in the tumour failure subgroup compared to one tumour in the complications subgroup. Mitotic figures indicate the ability of tumour cells to reproduce resulting in active intra-ocular growth, and they had been linked to high risk of metastasis before (Vrabec et al. 1991; Egan et al. 1998; Singh et al. 2001; Dendale et al. 2006). Monosomy 3 and gain of chromosome 8(q) were also frequently present in this series and resemble the distinct chromosomal alterations known as negative prognostic factors in primarily enucleated eyes containing UM (Prescher et al. 1994; & Sisley et al. 1997; White et al. 1998; Kiliç et al. 2005; Kiliç et al. 2006; van den Bosch et al. 2012). The tumours with monosomy 3 and/or gain of 8q, and high mitotic numbers could thus potentially also mark the subgroup of more aggressive or radio-resistant tumours. A total radiation dose of 50 Gray was found to be the optimal dose for killing most radiosensitive cells, yet spare the critical structures (Muller et al. 2005). Increasing the dose could potentially lower the number of cases with inadequate tumour control but could on the other hand also increase the number of secondary enucleations due to a painful blind eye. Until now, several regions on chromosome 8q have been suggested to harbour candidate oncogenes involved in UM such as MYC (Parrella et al. 2001) or DDEF1 (Ehlers & Harbour 2005). It is unknown whether these genes are involved in radiotherapeutic sensitivity as well. With the present follow-up, no increased risk of metastasis could be determined for patients in either secondary enucleation subgroup. This is contradictory to other reports and could be due to the small group of patients in this study, as well as the limited follow-up time.

One of the limitations of our study is the small group of patients who underwent secondary enucleation. As the incidence of secondary enucleation

after eye-conserving therapies is low, it is generally difficult to obtain data of a large group of secondarily enucleated patients. We are therefore cautious in drawing conclusions based on the group of patients who underwent secondary enucleation, and they provide, however, an insight in the histopathological and cytogenetical alterations of secondary enucleation specimens. As mentioned before, we cannot be sure about the timing of the alterations and which alterations were a result of fSRT. Also, FISH analysis on irradiated tumour tissue is difficult due to necrotic cells, but if sufficient DNA can be isolated from tumour cells, Multiplex Ligation dependent Probe Amplification (MLPA) or single nucleotide polymorphism (SNP) array analysis may provide data in these difficult cases (Vaarwater et al. 2012; Larsen et al. 2014). SNP array would also be suitable for the analysis of chromosomal regions with loss of heterozygosity, as previously reported by Lake et al. (2010).

In summary, we have found that tumour thickness indicates a high risk of secondary enucleation after fSRT. Mitotic figures and gain of chromosome 8q were frequently found in UM of enucleated eyes after previous fSRT and possibly indicate a more aggressive or radiation-resistant tumour. This is to our knowledge the first report with examination of cytogenetical and histopathological factors next to analysis of clinical risk factors for secondary enucleation after failed eye-conserving therapy and may thus serve as a starting point for further research. Future genetical research on biopsy material, taken before patients will be treated with fSRT, may provide us with more information about risk factors and possibly also differentiate the non-responding tumours from the responding ones.

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